

Homework 4 Solutions

HOMEWORK 4 SOLUTIONS

Problem 1.

Part A. The key observation is that $M^* := \text{AES}_K^{-1}(0^{128})$ is a *fixed point* of the round function when the incoming chaining value is 0^{128} : both $B[i]$ and $C[i]$ collapse to 0^{128} . Because $C[0] = 0^{128}$ is fixed, processing M^* returns the chain to 0^{128} , so appending another copy of M^* repeats the same computation and yields the same final output.

Adversary $A(K)$:
 $M^* \leftarrow \text{AES}_K^{-1}(0^{128})$
 Return $(M^*, M^* \| M^*)$

Correctness: $H_K(M^*) = 0^{128}$. Since M^* is a single 128-bit block, $m = 1$. Starting from $C[0] = 0^{128}$:

$$B[1] = \text{AES}_K(C[0] \oplus M^*) = \text{AES}_K(M^*) = 0^{128}, \quad C[1] = \text{AES}_K(B[1] \oplus M^*) = \text{AES}_K(M^*) = 0^{128}.$$

So $H_K(M^*) = C[1] = 0^{128}$.

Advantage analysis: $H_K(M^* \| M^*) = 0^{128}$. Now $m = 2$. Block 1 gives $C[1] = 0^{128}$ by the same computation above. Block 2 starts from this same chaining value 0^{128} , so the computation repeats identically:

$$B[2] = \text{AES}_K(C[1] \oplus M^*) = \text{AES}_K(M^*) = 0^{128}, \quad C[2] = \text{AES}_K(B[2] \oplus M^*) = \text{AES}_K(M^*) = 0^{128}.$$

So $H_K(M^* \| M^*) = C[2] = 0^{128} = H_K(M^*)$. Note $|M^*| = 128$ and $|M^* \| M^*| = 256$, so the two messages are distinct and lie in the domain of H . Therefore, for every key $K \in \{0, 1\}^{128}$ the adversary outputs a valid collision, so

$$\text{Adv}_H^{\text{cr}}(A) = 1.$$

Running time analysis. A is efficient as its running time is T_{AES} plus minor additional work.

Part B. Prepending M^* to any target M_1 does not change the final hash output. The reason is the fixed-point property from Part A: after processing M^* the chaining value resets to 0^{128} , which is also the starting value $C[0]$ used when hashing M_1 alone. The subsequent computation on M_1 is therefore identical in both cases.

Adversary $A(K, M_1)$:
 $M^* \leftarrow \text{AES}_K^{-1}(0^{128})$
 Return $M^* \| M_1$

$M_2 \in D$. $|M_2| = 128 + 128 = 256 = 2 \times 128$, which is a positive multiple of 128.

$M_2 \neq M_1$. $|M_2| = 256 \neq 128 = |M_1|$, so the two messages are distinct.

Advantage analysis: $H_K(M^* \| M_1) = H_K(M_1)$. We trace H_K on $M_2 = M^* \| M_1$. Block 1 (M^*), starting from $C[0] = 0^{128}$:

$$B[1] = \text{AES}_K(C[0] \oplus M^*) = \text{AES}_K(M^*) = 0^{128}, \quad C[1] = \text{AES}_K(B[1] \oplus M^*) = \text{AES}_K(M^*) = 0^{128}.$$

Block 2 (M_1), starting from $C[1] = 0^{128}$:

$$B[2] = \text{AES}_K(C[1] \oplus M_1) = \text{AES}_K(M_1), \quad C[2] = \text{AES}_K(B[2] \oplus M_1) = \text{AES}_K(\text{AES}_K(M_1) \oplus M_1).$$

Now trace H_K on M_1 alone, starting from $C[0] = 0^{128}$:

$$B[1] = \text{AES}_K(M_1), \quad C[1] = \text{AES}_K(\text{AES}_K(M_1) \oplus M_1).$$

Since $C[1] = 0^{128}$ in the two-block computation equals $C[0] = 0^{128}$ in the one-block computation, block 2 of the former and block 1 of the latter are processed from the same chaining value. All subsequent steps are therefore identical, giving

$$H_K(M^* \| M_1) = \text{AES}_K(\text{AES}_K(M_1) \oplus M_1) = H_K(M_1)$$

for every $K \in \{0, 1\}^{128}$ and every $M_1 \in \{0, 1\}^{128}$. As the game samples $K \leftarrow_{\$} \{0, 1\}^{128}$ and $M_1 \leftarrow_{\$} \{0, 1\}^{128}$, and for every possible outcome of such (K, M_1) we have $M_2 \in D$, $M_2 \neq M_1$, and $H_K(M_2) = H_K(M_1)$, FINALIZE returns 1 with probability 1, i.e.:

$$\text{Adv}_H^{\text{spr}}(A) = \Pr [\text{SPR}_H^A \Rightarrow 1] = 1.$$

Running time analysis. A is efficient as its running time is T_{AES} plus minor additional work.